



MASTER IN ASTROPHYSICS  
AND SPACE SCIENCE



TOR VERGATA  
UNIVERSITÀ DEGLI STUDI DI ROMA

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# Computational Fluid Dynamics

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

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January 23, 2026

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## Overview

### Physical Background

Fundamental Definitions

Basic Equations

Examples

### Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

### Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

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## Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions
  - small enough to ignore systematic variations, size below any characteristic scale length

$$\ell \ll \left| \frac{q}{\nabla q} \right|$$

- large enough to ignore discreteness noise, statistically sufficient number of particles

$$n\ell^3 \gg 1$$

- in collisional fluids, large enough to communicate local conditions, requires interactions

$$\ell \gg \lambda$$

## Describing Fluid Equations

- *Eulerian View*: cells of small volumes in fixed grid, flow is defined for all  $\mathbf{t}$  at position  $\mathbf{r}$  with rate of change  $\partial/\partial\mathbf{t}$  for any measurable quantity  $Q$
- *Lagrangian View*: follow particular fluid elements, comoving reference system with flow evaluated in each one via  $D/D\mathbf{t}$  as the rate of change
- *Translating Relation*: add convective derivative term to account for movement of fluid element

$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + \mathbf{u} \cdot \nabla Q$$

## Continuous Variables

### ■ *Primitive Fields:*

- velocity  $\mathbf{u}$  (vector)
- pressure  $p$  (scalar)
- density  $\rho$  (scalar)
- temperature  $T$  (scalar)

### ■ *Conserved Quantities:*

- mass  $M$  (scalar)
- momentum  $\mathbf{P}$  (vector)
- energy  $E$  (scalar)

### ■ *Derived Values:*

- viscosity  $\nu$  (scalar)
- vorticity  $\boldsymbol{\omega}$  (vector)

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## Frequentist View

- *With Prior Knowledge*: out of all  $n$  different but equally likely possible outcomes of an event, there are  $k$  that have a property  $A$  which is therefore measured with probability

$$P(A) = \frac{k}{n}$$

- *Without Prior Knowledge*: out of  $n$  independent measurements the outcome  $A$  occurs  $k$  times, defining

$$P(A) = \lim_{n \rightarrow \infty} \frac{k}{n}$$

## Bayesian View

- $P(A|B)$  quantifies the plausibility of an assumption  $A$  under known information given by  $B$
- *Kolmogorov Axioms* underpin the rules for calculating with probabilities
- $P(A, B) = P(A) P(B|A)$  quantifies the probability of measuring  $A$  and  $B$  simultaneously, and from assuming  $P(A, B) = P(B, A)$  we find

### Bayes' Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

## Continuous Distributions

■ *Cumulative Distribution Function:*  $F(x) = P(X \leq x)$

■ *Probability Density Function:*  $f(x) = \frac{d}{dx} F(x)$

$$P(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a)$$

■ *Normalization:*  $1 = \int_{-\infty}^{+\infty} f(x) dx$   $\lim_{x \rightarrow -\infty} F(x) = 0$   $\lim_{x \rightarrow +\infty} F(x) = 1$

## Expectation Values

$$E[h(x)] = \int_{-\infty}^{+\infty} h(x) f(x) dx$$

■ *Mean* or first raw moment:  $\mu = E[X] = \int_{-\infty}^{+\infty} x f(x) dx$

■ *Variance* or second raw moment:  $\sigma^2 = E[(X - E[X])^2] = \int_{-\infty}^{+\infty} (x - E[x])^2 f(x) dx$

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### ■ Frameworks:

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

### ■ Regimes:

- *Plasma*: Magnetohydrodynamics (MHD)
- *Degeneracy*: Quantum Effects
- *Collisionless*: Initial Distributions

### ■ Applications:

- Stellar Evolution
- Accretion Disks
- Cosmic Filaments

# Thank You!

- C. Clarke & B. Carswell, *Principles of Astrophysical Fluid Dynamics* (Cambridge University Press) 2007
- L. A. Barba, *CFD Python: 12 Steps to Navier-Stokes*<sup>1</sup> 2022
- P. Mocz, *Medium*<sup>2</sup> 2020-2025

Resources on GitHub: [here](#)